

SALM: Social Arabic Language Model

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Abstract

Arabic NLP faces a fundamental mismatch between training and deployment conditions. While most language models are trained predominantly on Modern Standard Arabic (MSA), real-world communication occurs largely in spoken dialects. To bridge this gap, we leverage social media content from Reddit, Twitter, and YouTube as a scalable proxy for spoken Arabic. We construct a 1.04-billion-token pretraining corpus and train SALM, a compact 336M-parameter decoder-only language model pre-trained via next-token prediction alone without masked language modelling, dialect labels, or task-specific signal. Despite an order-of-magnitude smaller training corpus than existing dialectal PLMs and a generative training objective that provides no direct classification signal, SALM outperforms MARBERT, explicitly pre-trained on one billion dialectal tweets, on MADAR-26 dialect identification by 4.2 accuracy points and 4.8 macro-F1 points, and leads all seven evaluated baselines outright on both MADAR-26 and MADAR-9. Fine-tuning SALM end-to-end further raises its own MADAR-26 accuracy from 44.9% to 57.0% (macro-F1 44.3% $\rightarrow$ 57.2%). Under the same frozen probe on binary Arabic sentiment (AJGT), SALM reaches 92.2% accuracy, the highest among evaluated Arabic PLMs, encoder and decoder alike, under a single-seed evaluation. We release the corpus and trained model to facilitate future research on Arabic spoken language modelling.

1 Introduction

Arabic is spoken by approximately 400 million native speakers across 22 countries, making it the fifth most widely spoken language in the world. This geographic breadth produces substantial linguistic diversity: the Arabic-speaking world spans from Morocco in the west to Oman in the east,

and each region has developed its own spoken dialect. Western varieties such as Moroccan Dar-ija blend Classical Arabic with Amazigh substrate vocabulary and French loanwords; Egyptian Arabic, the most widely understood supra-regional variety, serves as an informal spoken lingua franca; Gulf dialects preserve Bedouin phonological patterns; and Iraqi Arabic retains morphological features largely absent from MSA. Despite this diversity, the prestige written form, Modern Standard Arabic (MSA), is a single codified variety taught in schools and used across formal media throughout the Arab world.

Current NLP systems are poorly equipped for this diversity for two reasons. First, the data used to train large language models skews heavily toward written formal text: Wikipedia articles, news corpora, and digitised books are overwhelmingly MSA, systematically underrepresenting the spoken varieties that characterise everyday communication. Second, dialects are not simply noisy variants of MSA: they differ in morphology, phonology, lexicon, and syntax to a degree that can produce mutual unintelligibility, making an MSA-trained model a weak prior for dialectal text (Zaidan and Callison-Burch, 2014).

Social media platforms, Twitter, Reddit, and YouTube, offer a scalable and naturalistic solution. Hundreds of millions of Arabic speakers communicate daily in their native dialects on these platforms, generating a continuous stream of authentic dialectal text. We leverage this observation to construct SALM (Social Arabic Language Model), a 336M-parameter decoder-only causal language model pre-trained on a 1.04-billion-token corpus of Arabic social media text. Unlike encoder-based Arabic PLMs, SALM is trained solely with a causal language-modelling objective and uses no dialect supervision or task-specific pre-training signal. Its 1.04-billion-token corpus is an order of magnitude smaller in token count than the one-

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billion-tweet training set of MARBERT (Abdul-Mageed et al., 2021), and the entire training run fits on two consumer-class GPUs. Under a frozen linear-probe protocol, SALM outperforms MARBERT and every other evaluated baseline on the two most fine-grained MADAR granularities.

We summarize our contributions as follows:

- A 1.04-billion-token corpus of dialectal Arabic text curated and cleaned from Twitter, Reddit, and YouTube.
- SALM, a compact 336M-parameter decoder-only causal language model trained on this corpus via next-token prediction alone, with the pre-trained checkpoint publicly released.
- A frozen linear-probe and full fine-tuning evaluation on dialect identification (MADAR) and sentiment analysis (AJGT), demonstrating that SALM outperforms purpose-built dialectal encoders on fine-grained dialect identification despite a training corpus an order of magnitude smaller and without any classification signal.

2 Related Work

MSA-focused Arabic PLMs. Arabic NLP has long been dominated by models targeting Modern Standard Arabic. AraBERT (Antoun et al., 2020) extends masked language modelling to Arabic by pre-training on large MSA corpora drawn from news and Wikipedia, and ArBERT (Abdul-Mageed et al., 2021) follows a similar approach. These models achieve strong results on MSA benchmarks but do not directly address the dialectal diversity of spoken Arabic.

Dialect-aware Arabic PLMs. MARBERT (Abdul-Mageed et al., 2021) is pre-trained on one billion Arabic tweets and is the current state-of-the-art representation model for Arabic dialect tasks. CAMELBERT-DA (Inoue et al., 2021) is trained specifically on dialectal Arabic text and achieves competitive results on dialect identification. Both models are encoder-only (bidirectional BERT-style), which confers a structural advantage in classification tasks relative to decoder-only architectures. Crucially, both adopt a masked-language-modelling pretraining objective specifically designed to produce classification-ready representations.

Arabic generative models. AraGPT2 (Antoun et al., 2021) adapts GPT-2 (Radford et al., 2019) to Arabic using MSA text, demonstrating that decoder-only pre-training is viable for Arabic but limited to the formal written register. More recent large-scale efforts apply instruction-tuned GPT-style pre-training at the multi-billion-parameter scale, again with an MSA focus.

Dialectal language models. The closest work to ours in spirit is AlCLaM (Aldarmaki et al., 2024), a BERT-based encoder pre-trained from scratch on dialectal Arabic text collected from social media and evaluated on downstream discriminative tasks.¹ Unlike AlCLaM, SALM is a decoder-only causal language model: rather than a masked-language-modelling objective tailored to classification, SALM is trained solely via next-token prediction and is evaluated on downstream classification benchmarks, dialect identification and sentiment analysis, using both frozen and fine-tuned representations.

3 Dataset

We curated pretraining data from three social media platforms: Reddit, Twitter, and YouTube. All collected data is publicly available and was obtained using standard libraries and APIs in compliance with each platform’s terms of service. Table 1 reports corpus statistics; token counts are exact figures derived from the tokenised .npy shards rather than character-based estimates.

Corpus	Documents	Tokens	Share
Twitter	202,700,438	730M	70.5%
Reddit	4,670,468	240M	23.2%
YouTube	25,100	66M	6.3%
Total	207M+	1.04B	100%

Table 1: Pretraining corpus statistics (exact BPE token counts from tokenised shards; Twitter document count is the full pain/Arabic-Tweets dataset, tokens after quality filtering).

Reddit. We collect from 57 Arabic-speaking subreddits spanning the Arab world, with the largest communities centred on Egypt (*r/CAIRO*, *r/Egypt*, *r/ExEgypt*) and Saudi Arabia (*r/saudiArabia*, *r/SaudiForSaudis*). The corpus comprises 4,670,468 documents after cleaning (24.7% of 18.9M raw rows), with a median

¹AlCLaM reports 3.4M training sentences (no token count); our 207M+ raw documents (Table 1) are roughly sixty times more.

length of 129 characters. 88% of documents are comments rather than top-level posts, reflecting the conversational register of the data. Posts span from January 2009 to June 2026, providing temporal diversity.

Twitter. We use the pain/Arabic-Tweets dataset from HuggingFace, containing 202.7M Arabic tweets. Tweets are short (median 63 characters, 12 words) and uniformly high-quality in terms of Arabic script content (Arabic character ratio = 1.0 after cleaning). Despite the large raw document count, the tokenised corpus yields 730M tokens after quality filtering, reflecting the filtering of very short or non-dialectal tweets.

YouTube. We collect transcripts of Arabic-language YouTube videos, primarily movies, series, and podcasts, yielding 25,100 unique videos after exact- and near-duplicate removal. Transcripts are substantially longer than social media posts (median 3,416 characters, 684 words), providing the model with extended discourse context absent from Twitter and Reddit.

Appendix B (Figure 2) gives paraphrased examples of raw text from each platform, illustrating their register and dialect mix.

Dialect coverage. We characterise dialect distribution across all three corpora using camelbert², a BERT model fine-tuned on the NADI 2020 country-level Arabic dialect identification shared task (Abdul-Mageed et al., 2020), which classifies documents into 21 Arabic-speaking countries with no reject/unassigned option. We draw a random sample of 8,000 documents per corpus (fixed seed 42); YouTube transcripts, being substantially longer than tweets or Reddit posts, are first windowed into ~ 40 -word, tweet-length chunks so the classifier sees comparable-length input across all three corpora. We collapse the 21 country-level predictions into seven broad dialect regions used elsewhere in this paper (Egyptian, Levantine, Gulf, Iraqi, Maghrebi, Sudanese, Yemeni) plus an Other bucket for the small number of countries with no clean regional match. Table 2 reports the resulting region shares; every sampled document receives a regional assignment by construction. Egyptian Arabic (EGY) is the dominant variety on Reddit and YouTube (45.8% and 55.7% respectively), reflecting Egypt’s large online presence and media output. Gulf Arabic (GLF) in-

stead dominates on Twitter (48.6%), with Egyptian second (20.0%), consistent with high social media engagement across the Gulf region. Egyptian and Gulf content jointly account for 72–80% of sampled documents on every platform; Levantine, Maghrebi, and Iraqi Arabic are each present at low-single-digit-to-low-teen shares, while Sudanese and Yemeni Arabic remain under 5% on every platform (see Limitations).

Region	Reddit	YouTube	Twitter
EGY (Egyptian)	45.8%	55.7%	20.0%
GLF (Gulf)	26.7%	24.6%	48.6%
LEV (Levantine)	11.2%	6.8%	5.5%
MAG (Maghrebi)	7.0%	3.3%	10.3%
IRQ (Iraqi)	4.9%	4.5%	10.6%
YEM (Yemeni)	3.9%	4.8%	3.9%
SUD (Sudanese)	0.4%	0.2%	0.4%
Other	0.2%	0.1%	0.8%

Table 2: Dialect region share, 8,000-document random sample per corpus (seed = 42), collapsed from 21 camelbert-mix-did-nadi countries to 7 regions (0% unassigned).

3.1 Preprocessing

We conduct heavy preprocessing of the social media text prior to tokenisation. Social media text is characteristically noisy, exhibiting code-switching, non-Arabic script, URLs, diacritics, and punctuation-heavy formatting. Inspired by (El-Haj et al., 2024), we develop a multi-stage filter applied uniformly across all three sources: (1) removal of Latin characters and non-Arabic punctuation,³ (2) stripping of diacritics (tashkeel), (3) a minimum character-length gate (20 characters post-cleaning), (4) an Arabic content gate (at least 40% of non-whitespace characters must be Arabic script), and (5) exact- and near-duplicate removal for YouTube transcripts (MinHash, Jaccard threshold = 0.8). For Reddit, an additional 11.9M documents (62.9% of raw rows) were removed as punctuation-only or near-empty entries. The final output per document is a single cleaned line containing the social media post with no punctuation, diacritics, or non-Arabic tokens. Appendix A provides detailed per-platform cleaning statistics.

³This step also removes Arabizi (Arabic transliterated into Latin script) and code-switched Latin-script segments, since the pipeline has no transliteration component to map them back to Arabic script. We made this deliberate choice for simplicity and acknowledge its limitations (see Limitations).

²CAMEL-Lab/bert-base-arabic-camelbert-mix-did-nadi

4 SALM Model

4.1 Model Architecture

SALM follows the GPT-2 decoder-only Transformer architecture (Radford et al., 2019). Each block applies pre-layer normalisation, causal multi-head self-attention computed via scaled dot-product attention, and a position-wise feed-forward network with GELU activation (tanh approximation). A single layer normalisation is applied before the output projection. Token embedding and output projection weights are tied to reduce parameter count.

SALM has 24 transformer layers, 16 attention heads, and a model dimension $d_{\text{model}}=1024$, totalling 336M parameters. The model uses a context length of 512 tokens and a vocabulary of 32,000 BPE subword units (Section 4.2).

4.2 Tokenizer

Arabic NLP systems typically rely on tokenizers trained on MSA-heavy corpora, which undersegment dialectal words absent from their training distribution. We train a custom BPE tokenizer on a subset of the pretraining corpus (Reddit and YouTube), ensuring the vocabulary reflects the dialectal and social media register of the data.

Design. We use a Whitespace pre-tokenizer rather than the ByteLevel encoding common in multilingual models. ByteLevel maps each byte to a printable ASCII character, distorting Arabic codepoints and producing unintuitive subword boundaries; Whitespace splitting instead preserves Arabic character identity and aligns splits with dialectal morphological units. No normaliser is applied beyond what has already been performed in preprocessing (Section 3.1); further normalisation risks collapsing dialectal character variants that carry morphological information. Subword continuation pieces are marked with a ## prefix, and the decoder uses a WordPiece-style strategy (rather than the ByteLevel BPE decoder) to correctly reinsert spaces on decode.

Vocabulary. The vocabulary comprises 32,000 BPE types trained with minimum merge frequency 2, plus five special tokens (<pad>, <unk>, <bos>, <eos>, <mask>). Of the 32,000 types, 31,555 (98.6%) contain at least one Arabic character; only 5 types contain Latin characters, reflecting the Arabic-script-only training data. Zero diacritic-bearing types appear in the vocabulary, consis-

tent with diacritic removal in preprocessing. Subword pieces (marked ##) account for 5,464 types (17.1%); the majority of vocabulary entries are therefore whole words, appropriate for the short utterances that dominate social media text.

Fertility. Across Egyptian, Levantine, and Iraqi samples, fertility is approximately 1.0–1.1 tokens per whitespace-delimited word: common dialectal words such as *mish* (not), *wallah* (by God), and *btihki* (you speak) each map to a single token. Gulf and Maghrebi varieties show marginally higher fertility due to a small number of characters absent from the training data, which are mapped to <unk>. Latin-script text (Arabizi, code-switched English) maps entirely to <unk> by design, as the pretraining corpus is Arabic-script only. Encode–decode round-trips are lossless for the three major varieties tested; any mismatch in Gulf or Maghrebi samples is attributable to a single <unk> substitution per sentence with no subword leakage.

4.3 Training Setup

SALM is trained on two L40S GPUs (48 GB each) with an effective batch size of 131,072 tokens per gradient step, achieved via micro-batching with gradient accumulation (micro-batch $B=8$, sequence length $T=512$, accumulation steps=16 across 2 GPUs). The optimiser is fused AdamW ($\beta_1=0.9$, $\beta_2=0.95$, $\varepsilon=10^{-8}$, weight decay=0.1) with gradient norm clipping at 1.0. The learning rate follows a linear warmup over 315 steps (approximately 3% of the planned training schedule) followed by cosine annealing to a minimum of 6×10^{-5} ; the peak learning rate is 6×10^{-4} . Training runs in bfloat16 mixed precision with `float32_matmul_precision='high'`.

SALM is evaluated at the checkpoint corresponding to step 18,950 (2.48B training tokens consumed). Validation cross-entropy loss at this checkpoint is 4.36 on a held-out 10% split of the pretraining corpus (≈ 100 M tokens), and the training curve shows no plateau at this point, indicating the model would benefit from continued training.

5 Experiments

5.1 Evaluation Protocol

We evaluate SALM using two protocols. The primary evaluation is a **frozen linear probe**: the backbone is held fixed and a single linear classification layer is trained on mean-pooled, L2-normalised sentence embeddings extracted from

the frozen model. A frozen probe directly tests whether task-relevant structure is linearly decodable from pretrained representations (Alain and Bengio, 2017), without allowing the backbone to restructure in response to task labels. It also places encoder and decoder architectures on equal footing: bidirectional pooling confers an advantage under fine-tuning, but both reduce to a fixed embedding function under the probe, making performance differences attributable to pretraining corpus effects rather than architectural inductive biases.

The probe is trained with class-weighted cross-entropy (AdamW, lr=1e-2, weight_decay=1e-4) for up to 20 epochs, selecting the epoch with the best accuracy on a held-out validation split. The train/validation/test split used for each task is specified in the corresponding subsection below.

5.2 Dialect Identification

We evaluate on four granularities derived from the MADAR parallel corpus (Bouamor et al., 2019) (12,000 BTEC sentences, 26 city-dialect varieties): **MADAR-2** (binary, MSA vs. dialect), **MADAR-6** (six broad regions: EGY, LEV, GLF, IRQ, MAG, MSA), **MADAR-9** (nine regions: Nile Egypt, Nile Sudan, Levant, Iraq, Gulf, Yemen, Libya, Maghreb, MSA), and **MADAR-26** (26 city-level varieties). The parallel corpus is unstacked so that each non-empty (sentence, city) cell becomes one labelled example, yielding 112,000 examples total. Because MADAR is a parallel corpus, the same underlying BTEC sentence is translated into all 26 city dialects; we therefore partition examples using MADAR’s official per-sentence train/dev/test assignment (95,600 / 11,200 / 5,200 examples) rather than a fresh per-example split, so that different-dialect translations of the same sentence cannot cross the train/test boundary.⁴ Model selection (epoch/checkpoint choice, Section 5.1) uses the dev split; the held-out test split is scored exactly once per model, with the selected weights.

We compare SALM against six Arabic and multilingual PLM baselines. Majority-class baselines on the test split are: MADAR-2 96.2%, MADAR-6 26.9%, MADAR-9 23.1%, MADAR-26 3.8%.

⁴The test split fixes 200 sentences/city ($200 \times 26 = 5,200$), so it is not class-balanced across the coarser MADAR-2/6/9 label sets, affecting their majority baselines below.

5.3 Sentiment Analysis

We evaluate on the AJGT (Arabic Jordanian General Tweets) dataset (Alomari et al., 2017), a binary sentiment corpus of 1,800 tweets (positive or negative) with a balanced class distribution of 900 examples per class. AJGT has no official train/test partition, so, unlike MADAR (Section 5.2), we use the same frozen probe, embedding extraction, hyperparameters, and a stratified 80/20 train/test split (seed=42) as described in Section 5.1.⁵

5.4 Full Fine-tuning

To assess SALM’s representational capacity beyond what a frozen linear probe reveals, we additionally fine-tune SALM end-to-end on each MADAR variant. A linear classification head is placed on the mean-pooled final hidden states; all backbone weights receive gradients. We use AdamW with a backbone learning rate of 2×10^{-5} and a head learning rate of 1×10^{-3} , cosine-decayed with 10% linear warmup over the first epoch. Training runs for up to 10 epochs with early stopping on validation accuracy (patience=3), using the same official MADAR train/dev/test split as Section 5.2 throughout, so results are directly comparable to the frozen-probe evaluation.

5.5 Training Evaluation

We monitor training progress by evaluating validation loss every 475 steps on a held-out 10% split of the pretraining corpus (approximately 100M tokens). Checkpoints are saved at the same cadence, enabling post-hoc selection of the best checkpoint for downstream evaluation. SALM’s training loss declines from 10.6 at initialisation to 4.6 at step 18,950 with no observed plateau, indicating the model is likely to benefit from continued training.

6 Results

6.1 Dialect Identification

Table 3 reports accuracy and macro-F1 across all four MADAR variants for the frozen linear probe.

SALM leads on fine-grained dialect identification. SALM achieves 44.9% accuracy and 44.3% macro-F1 on MADAR-26, the best score of any evaluated model on both metrics, ahead of the next-best accuracy (CAMEL, 42.3%) and

⁵AJGT may overlap with the pain/Arabic-Tweets pre-training data, though this is unlikely given AJGT’s narrow construction (a small, curated set of Jordanian tweets).

Model	Accuracy (%)				Macro-F1 (%)			
	M-2	M-6	M-9	M-26	M-2	M-6	M-9	M-26
<i>Multilingual PLMs</i>								
mBERT	88.4	50.8	45.2	31.0	65.4	49.5	41.0	29.7
LaBSE	87.1	44.9	39.1	23.4	63.5	43.4	34.2	22.0
<i>MSA-focused Arabic PLMs</i>								
AraBERT	94.2	64.6	59.5	37.8	76.6	64.0	54.9	36.9
ArBERT	91.8	64.3	59.8	38.5	71.3	62.6	54.3	37.5
<i>MSA+Dialect Arabic PLMs</i>								
MARBERT	92.0	70.9	63.9	40.7	71.5	68.3	58.3	39.5
CAMEL	94.5	68.0	63.2	42.3	76.9	66.7	57.5	41.4
<i>This work (frozen linear probe)</i>								
SALM	93.0	69.5	65.1	44.9	73.5	67.5	59.5	44.3

Table 3: Dialect identification via frozen linear probe, scored once on the official held-out test split (Section 5.2). M-2/6/9/26 = MADAR granularity. **Bold** = best per column.

the next-best macro-F1 (MARBERT, 39.5%), despite never being trained on any classification signal. SALM also leads MADAR-9 (65.1% accuracy, 59.5% macro-F1), narrowly ahead of MARBERT (63.9%, 58.3%). At the coarser MADAR-2 and MADAR-6 granularities SALM is competitive but not best: CAMEL leads MADAR-2⁶ (94.5% accuracy, 76.9% macro-F1) and MARBERT leads MADAR-6 (70.9% accuracy, 68.3% macro-F1).

Arabic-specific pre-training dominates over multilingual scale. LaBSE and mBERT are the weakest baselines (23.4% and 31.0% accuracy at MADAR-26), confirming that Arabic-specific pre-training contributes far more to dialectal representation quality than multilingual scale alone. The gap between mBERT and SALM at MADAR-26 is 13.9pp; against LaBSE it is 21.5pp.

6.2 Sentiment Analysis

Table 4 reports accuracy and macro-F1 on AJGT. The dataset is perfectly balanced (900 per class), so the majority baseline is 50.0% and accuracy and macro-F1 are numerically near-identical throughout.

SALM achieves the highest sentiment accuracy among evaluated PLMs. SALM achieves 92.2%, the highest accuracy of any evaluated model, ahead of the strongest encoders ArBERT (91.9%) and CAMELBERT-DA (91.7%). The largest gap is between SALM and MARBERT (87.5%, -4.7pp), a model pre-trained on one billion dialectal tweets with a classification-oriented MLM objective and training data an order of mag-

⁶Every model falls below the 96.2% MADAR-2 majority baseline: the class-weighted probe trades raw accuracy for minority (MSA) recall, so macro-F1 ($\approx 49\%$ for the majority baseline vs. 63.5–76.9% here) is the more informative metric.

Model	Acc (%)	Macro-F1 (%)
<i>Multilingual PLMs</i>		
mBERT	76.4	76.3
LaBSE	88.1	88.0
<i>MSA-focused Arabic PLMs</i>		
AraBERT	86.9	86.9
ArBERT	91.9	91.9
<i>MSA+Dialect Arabic PLMs</i>		
MARBERT	87.5	87.5
CAMELBERT-DA	91.7	91.7
<i>This work (frozen linear probe)</i>		
SALM	92.2	92.2

Table 4: AJGT sentiment analysis via frozen linear probe, single stratified 80/20 split and seed (Section 5.3). Majority baseline: 50.0%. **Bold** = best overall.

nitude larger. This result is notable for a causal decoder-only model: SALM uses left-to-right attention with a purely generative training objective, yet attains the top accuracy among the bidirectional encoder baselines evaluated here. These results reflect a single train/test split and a single random seed; we have not established their significance across seeds, nor verified the absence of overlap between AJGT and the pretraining corpus (Limitations).

6.3 Fine-tuning Results

Table 5 contrasts full fine-tuning with the frozen probe for SALM. Fine-tuning yields consistent and substantial gains over SALM’s own frozen probe across all MADAR granularities: accuracy improves by +4.3pp on MADAR-2, +11.6pp on MADAR-6, +13.9pp on MADAR-9, and +12.1pp on MADAR-26, with macro-F1 gains of a similar magnitude (+11.6 to +15.2pp). Gains are smallest on the coarse MSA-vs-dialect dis-

tion (MADAR-2) and largest at the regional and city-level granularities, consistent with the frozen probe being most constrained where a linear head cannot compensate for causal mean-pooling’s weaker cross-token aggregation relative to a bidirectional encoder’s pooled representation.

We report MARBERT’s and CAMEL’s frozen-probe scores in Table 5 for context only. Because we do not fine-tune these baselines under an identical protocol (same optimiser schedule, dev-based early stopping, single test evaluation), Table 5 supports a claim about SALM’s own gain from fine-tuning, not a comparison against fine-tuned baselines.

7 Discussion

Small corpus, generative objective, competitive performance. SALM’s 1.04B-token corpus is an order of magnitude smaller than the one billion tweets used to pretrain MARBERT, an encoder-only model with a masked-language-modelling objective specifically designed to produce classification-ready representations; SALM instead uses only causal next-token prediction and no dialect supervision. Despite this, SALM outperforms MARBERT on MADAR-26 accuracy (44.9 vs. 40.7, +4.2pp) and macro-F1 (44.3 vs. 39.5, +4.8pp), and achieves the highest accuracy among evaluated PLMs on AJGT sentiment. This suggests that corpus quality and register relevance matter more than corpus scale or pretraining objective when the data adequately covers the target linguistic domain, and confirms that social media content is a high-quality, scalable proxy for spoken Arabic.

Fine-tuning further closes the architectural pooling gap. Even though SALM already leads the frozen-probe comparison at MADAR-9 and MADAR-26 (Section 5.2), full fine-tuning still yields large further gains over its own frozen-probe score: MADAR-26 accuracy rises from 44.9% to 57.0% (+12.1pp), and macro-F1 from 44.3% to 57.2% (+12.9pp). This is consistent with mean-pooled causal embeddings being a structurally weaker classification-ready representation than a bidirectional encoder’s pooled embedding: even when SALM’s frozen representations already contain enough dialectal signal to lead the frozen-probe comparison, allowing the backbone to adapt end-to-end still recovers substantial additional accuracy. We do not have a matched fine-tuned base-

line under an identical protocol to test whether an encoder would show a comparably sized gain (Section 5.4).

The corpus dialect distribution shapes representation quality. The dialect distribution in Table 2 shows that Maghrebi Arabic constitutes 7.0% of the Reddit sample and 3.3% of the YouTube sample, and Sudanese Arabic under 0.4% of every platform. MADAR-26 includes several Maghrebi city labels (Tunis, Rabat, Sfax, etc.), and macro-F1 penalises underperformance on these and other minority classes. SALM’s macro-F1 lead over MARBERT at MADAR-26 (44.3 vs. 39.5, +4.8pp) suggests that the multi-platform corpus composition provides useful dialect coverage for minority-class recall, despite the Egyptian/Gulf skew documented in Table 2.

Qualitative generation. Figure 1 shows representative completions from SALM across four dialectal varieties (full continuations in Appendix C, Figure 3), each preserving variety-specific lexical items absent from MSA: Egyptian *يا ريت/احنا/ده*, Levantine *سبحني/مافي احد*, Iraqi *ماكو/اكو* (the most distinctive Iraqi marker), and Maghrebi *والو/كلين* (the latter borrowed from Amazigh, exclusive to Maghrebi Arabic). This qualitative evidence complements the linear probe results: the dialectal knowledge the probe captures reflects genuine structure encoded during pretraining, not an artefact of probe training.

Sentiment representations generalise from dialectal pretraining. The AJGT result (Section 5.3) extends the dialect identification finding: social media pretraining on a compact corpus induces representations broadly useful for Arabic sociolinguistic tasks, not only for dialect distinctions.

8 Conclusion

We introduced SALM, a compact 336M-parameter decoder-only language model pre-trained on a 1.04-billion-token corpus of dialectal Arabic social media text from Twitter, Reddit, and YouTube, using only a causal next-token-prediction objective, no masked language modelling, no dialect labels, no task-specific signal. Despite a training corpus an order of magnitude smaller than existing dialectal Arabic PLMs, SALM outperforms MARBERT and every other evaluated baseline on MADAR-9 and MADAR-26 dialect identification under a frozen linear probe, by 4.2 accuracy points and

Model	Accuracy (%)				Macro-F1 (%)			
	M-2	M-6	M-9	M-26	M-2	M-6	M-9	M-26
SALM _{probe}	93.0	69.5	65.1	44.9	73.5	67.5	59.5	44.3
SALM _{finetune}	97.3	81.1	79.0	57.0	85.2	79.1	74.7	57.2
MARBERT _{probe}	92.0	70.9	63.9	40.7	71.5	68.3	58.3	39.5
CAMEL _{probe}	94.5	68.0	63.2	42.3	76.9	66.7	57.5	41.4

Table 5: Full fine-tuning vs. frozen probe on MADAR for SALM. **Bold** = best SALM row per column. MARBERT/CAMEL frozen-probe scores are shown for context only, not fine-tuned comparison (Section 5.4).

Egyptian	Levantine	Iraqi	Maghrebi
في ناس بتقول ان مصر كانت دولة علمانية بس احنا بنعتبر ده كلام فارغ لان عندنا قانون <i>“Some people say Egypt was a secular state, but we think that’s empty talk because we have a law...”</i>	والله هاد الشي صار معي امبارح الساعة بالليل بس مافي احد بالبيت يارب ارزقني <i>“By God, this thing happened to me yesterday, at night but no one was home. Lord, provide for me...”</i>	صديقي شكرواكو كلشي تمام وين اكو احد يدزلي مسج بالواتساب حت لو من رقم غريب <i>“My friend, what’s up, everything’s fine — where is there anyone who’d send me a WhatsApp message, even from an unknown number?”</i>	واش راك كيفاش حالك الله يستر وين كين والوانت مش محتاج حد يحضنك <i>“How are you, how are you doing, may God protect you. Where is there anything? You don’t need anyone to hold you...”</i>

Figure 1: SALM completions across four dialectal varieties (temperature=0.85, top- k =50, repetition penalty=1.2). **Bold** = prompt; plain text = model continuation. Excerpted from the full prompts and continuations in Appendix C (Figure 3).

4.8 macro-F1 points at MADAR-26. Fine-tuning SALM end-to-end raises its own MADAR-26 accuracy to 57.0% (macro-F1 57.2%), a further gain of 12.1 accuracy points over its frozen probe. Under the same frozen probe on binary Arabic sentiment (AJGT), SALM achieves 92.2% accuracy, the highest among evaluated Arabic PLMs.

These results establish three conclusions. First, social media content is a high-quality and scalable training signal for dialectal Arabic representation learning even at modest corpus scale, with the induced representations generalising to sentiment analysis without any task-specific adaptation. Second, a decoder-only generative model pre-trained on social media can already match or exceed, under a frozen probe, encoder models purpose-built for classification, and fine-tuning yields substantial further gains on top of this. Third, even where SALM’s frozen-probe representations already lead the comparison, fine-tuning still recovers a large further improvement, consistent with mean-pooled causal embeddings being a structurally weaker classification-ready representation than a bidirectional encoder’s pooled embedding.

We release the pre-trained SALM model and evaluation code to support future work on Arabic dialect representation, generation, and zero-shot transfer.

Limitations

Architectural mismatch for classification. SALM is a causal (decoder-only) language model; representations are extracted by mean-pooling over the causal context, where each token attends only to preceding tokens. This is structurally weaker than bidirectional BERT-style pooling for classification tasks, so the frozen-probe results likely underestimate SALM’s representational capacity even where SALM already leads the frozen-probe comparison (Section 5.2), as the fine-tuning results in Section 5.4 demonstrate.

Corpus geographic bias. Egyptian Arabic dominates Reddit (45.8%) and YouTube (55.7%); Gulf Arabic dominates Twitter (48.6%). Egyptian and Gulf content jointly account for 72–80% of sampled documents on every platform (Table 2), while Sudanese Arabic represents under 0.4% and Yemeni Arabic under 5% of every corpus, a significant underrepresentation relative to their speaker populations in the Arab world. This skew may affect SALM’s representations of Sudanese and Yemeni dialects, both of which have dedicated labels in MADAR-9 and MADAR-26.

Sentiment evaluation robustness. The AJGT sentiment results (Section 5.3) reflect a single train/test split and a single random seed; we have not run a significance test (e.g. across multiple seeds, or a paired test such as McNemar’s) to es-

establish whether the reported margins over baselines are reliable rather than an artefact of this particular split. We have also not verified the absence of exact- or near-duplicate overlap between AJGT and the pretraining corpus. Because SALM’s pretraining objective is next-token prediction and never sees AJGT’s sentiment labels, such an overlap could at most make SALM’s representations unusually well-tuned to the surface form of specific tweets, not leak label information the way a supervised train/test overlap would, but this has not been empirically confirmed. Both checks are needed before drawing strong conclusions from the sentiment results.

Arabizi and code-switched text. Our preprocessing pipeline (Section 3.1) removes non-Arabic-script text, which excludes Arabizi (Arabic transliterated into Latin script) and code-switched segments common in real social media use. We have not measured what fraction of the raw scraped data this affects, so the corpus and the resulting model should be understood as covering Arabic-script dialectal text specifically, not the full range of scripts in which Arabic speakers communicate online.

Evaluation scope. This paper focuses on dialect identification and binary sentiment analysis. SALM has not been evaluated on MSA NLU tasks (NER, question answering), text generation quality (perplexity, human evaluation), or cross-lingual transfer. The sentiment evaluation uses dialectal Twitter data (AJGT) and does not cover MSA sentiment benchmarks. Whether social media pretraining benefits or harms MSA task performance remains an open question for future work.

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A Pretraining Corpus Statistics

Tables 6 and 7 provide detailed per-platform statistics after all filtering and deduplication steps.

Source	Example (paraphrased)
Twitter (Egyptian)	تسمع اغاني تامر حسني وانت بت رسم؟ دي احلى حاجه ممكن تعملها "Listening to Tamer Hosny while you draw? That's the loveliest thing you could be doing."
Twitter (Gulf)	الله يسعدك، بس ليش اخذت هالفكره؟ ابي افهم امتي كانت اخر مره زرتهم "God grant you happiness — but why did you get that idea? I want to understand: when was the last time you visited them?"
Reddit (r/cairo, Egyptian)	ابويا رجع من الخليج معاه شويه فلوس وعلايز يستثمرهم بس مالوش علاقات هنا. لو حد عنده افكار مشاريع كويسه يقولي "My dad just got back from the Gulf with some savings and wants to invest them, but he has no connections here — if anyone has good business ideas, let me know."
Reddit (r/jordan, Levantine)	سؤال للمتزوجين: قديش صرفتوا عالعرس كله؟ وشو الجزء يلي كلفكم اكثر شي؟ "A question for married folks: how much did the whole wedding cost you, and which part was the most expensive?"
YouTube (Egyptian, drama)	يا بنت عمي، انا مش قادره انسى اللي عمله فيا صوره امي واقفه قدامي طول الوقت، وحاسه اني لازم اخذ حقي منه "Cousin, I really can't forget what he did to me. My mother's face is in front of me all the time, and I feel like I have to get justice from him."
YouTube (Gulf, drama)	ما كنا نبغي نتخلص من ابوهم الظالم، بس الحمد لله مات وارتخنا منه الحين نقدر نعيش هني بخير، وننعم بالراحه والسعاده "We didn't wish their harsh father dead, but thank God he passed and we're free of him. Now we can live here well, and enjoy peace and happiness."

Figure 2: Representative examples of raw pretraining text by source. Examples are paraphrased from real posts, comments, and transcripts in our corpus rather than quoted verbatim (privacy; platform redistribution terms): wording and specifics are altered while preserving the original dialect, register, and topic. Twitter posts are short and topically varied; Reddit posts are longer and conversational; YouTube transcripts reflect the spoken dialogue register of movies, series, and podcasts.

	Twitter	Reddit	YouTube
Raw documents	202.7M	18.9M	26,189
After filtering	202.7M	4.67M	25,100
Retention rate	,	24.7%	95.8%
Median length (chars)	63	129	3,416
Median length (words)	12	,	684
Max length (chars)	280	,	,
Total tokens (BPE)	730M	240M	66M
Corpus share	70.5%	23.2%	6.3%
Arabic char ratio	1.00	>0.40	>0.40
Date range	,	Jan 09–Jun 26	,
% comments	,	88%	,
Unique subreddits	,	57	,

Table 6: Per-platform corpus statistics after cleaning and deduplication. Twitter raw/retention figures are not tracked separately (full pain/Arabic-Tweets dataset used). YouTube figures are unique videos after MinHash near-deduplication (Jaccard threshold = 0.8).

B Raw Corpus Examples

C Qualitative Generation Examples

Filtering step	Reddit	YouTube
Raw input	18,939,794	26,189
Punct.-only / near-empty	−11,916,915	,
Below min. length	−2,351,651	,
Non-Arabic script	−760	,
Exact duplicates	,	−222
Near-dups (MinHash)	,	−858
After filtering	4,670,468	25,100
Retention rate	24.7%	95.8%
<i>Character-level cleaning (Reddit)</i>		
Raw characters	4,027,096,974	
After cleaning	1,021,920,741	
Character retention	25.4%	
English words removed	501,067,877	
URLs removed	13,341	
Quranic chars stripped	241,940	

Table 7: Cleaning pipeline statistics for Reddit and YouTube; Twitter required no document-level filtering (Arabic char ratio = 1.0). Reddit’s low retention (25.4%) reflects aggressive removal of Latin-script and punctuation-heavy content.

Dialect	Prompt (bold) + generation (plain)
Egyptian	في ناس بتقول ان مصر كانت دوله علمانيه بس احنا بنعتبر ده كلام فارغ لان عندنا قانون وقانونا يسمح لنا بحمل السلاح ضد الرصاص مع اني في ناس بيتقولوا ان مصر لا تستطيع تحمل السلاح من غير السلاح ده يا ريت ما حدا يعلق عا صوت الموسيقى <i>"Some people say Egypt was a secular state, but we think that's empty talk because we have a law allowing us to bear arms , though some say Egypt can't carry arms without those weapons. I just wish nobody would comment on the music."</i>
Levantine	والله هاد الشي صار معي امبارح الساعة بالليل في صوت واحد قاعد عند الباب بس مافي احد بالبيت يارب ارزقني وانت خير الرازقين اللهم بشرني بما يسرني مما يخفي لي الفرح في الجاي يارب سخري جنود الارض وملائكة السماء <i>"By God, this thing happened to me yesterday , at night there was a sound at the door but no one was home. Lord, provide for me for You are the best of providers; give me good news of what hides joy ahead; Lord, enlist for me the soldiers of earth and the angels of heaven."</i>
Iraqi	صديقي شكو ماكو كلشي تمام وين اكو احد يدزلي مسج بالواتساب حت لو من رقم غريب لا صوت زين ولا مسج ليش هالحركه انا سمعت صوت انفجار قوي بس ما ركزت كنت نايمه فيه ولا شي <i>"My friend, what's up , everything's fine. Where is there anyone who'd send me a WhatsApp message, even from an unknown number? Not a good word, not a message. Why this silence? I heard a huge explosion but didn't pay attention , I was asleep, nothing."</i>
Maghrebi	واش رالك كيفاش حالك الله يستر وين كين والو انت مش محتاج حد يحضنك كل ما تقع او تنوجع اكثر من صوت افكارك اللي بتحسه يحسنلك جوده الحياه ويخليك تحس بالامان وانت قاعد في البيت <i>"How are you, how are you doing , may God protect you. Where is there anything? You don't need anyone to hold you every time you fall or hurt , just the sound of your own thoughts makes life feel better and gives you a sense of safety sitting at home."</i>

Figure 3: Dialectal completions from SALM (temperature=0.85, top- k =50, repetition penalty=1.2). Prompt in **bold**; continuation in plain text; dialect-specific items discussed in the text.